858.625.1885

rina.s.kim@gmail.com

San Diego, CA

https://www.linkedin.com/in/irina-s-kim/

IK

IRINA KIM

Search Committee Members

Halıcıoğlu Data Science Institute (HDSI)

University of California, San Diego

To:

RE: Faculty Position in AI Systems & Security

Dear Search Committee Members,

I am writing to formally apply for the faculty position within the Halıcıoğlu Data Science Institute (HDSI) at UC San Diego. As an IEEE Senior Member with a Ph.D. ABD in Applied Mathematics and 20 years of industrial leadership at global technology leaders such as Qualcomm and Samsung, my career has been dedicated to the mathematical and systemic foundations of Physical AI.

My research and professional contributions center on the "Full Stack of Pixels"—the intersection of hardware-aware machine learning, computer vision, and systems security. I hold 10 US patents in areas including stereo vision, ISP pipelines, and deep learning compression, reflecting a career-long commitment to advancing the state of image signal processing. At Qualcomm, I spearheaded the shift toward Bayer-domain Transformers and Latent Diffusion models, defining the requirements for next-generation NPU architectures to support these workloads.

Beyond technical contributions, I have a deep interest in the security and integrity of vision systems. My work on Under-Display Camera (UDC) technology and bit-accurate simulation environments addresses critical challenges in how AI systems perceive and process sensitive visual data. I am eager to bring this industrial perspective to HDSI to mentor the next generation of researchers and collaborate on the security of AI-driven perception systems.

I look forward to the opportunity to discuss how my blend of theoretical depth and production-grade engineering can contribute to the mission of UC San Diego.

Sincerely, Irina Kim.